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DETAILED SPECIFICATION

BURSTER, PROJECTILE - M54A1

This specification is approved for use by the U.S. Army Armament Research, Development and Engineering Center (ARDEC), and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers components, loading, assembling and packing of the M54A1 projectile burster.

2. APPLICABLE DOCUMENTS.

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to assure the completeness of this list, document users are cautioned that they must meet all requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to the Commander, U.S. Army ARDEC, ATTN: AMSRD-AAR-QES-E, Picatinny, NJ 07806-5000 or emailed to ardec-stdzn@pica.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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2.2 Government documents.

2.2.1 Specification, standards and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1168	-	Ammunition Lot Numbering and Ammunition Data Card
MIL-STD-1916	-	DOD Preferred Method of Acceptance of Product

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications below form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER (ARDEC) DRAWINGS

9238839	-	PROJECTILE BURSTER M54A1 ASSEMBLY
9238840	-	DISC
9238841	-	CLOSURE
9238842	-	BURSTER TUBE
9238843	-	BASE PLUG
9241806	-	COMPOSITION B-5

(Copies of these drawings may be requested online at Drawing-Request@pica.army.mil or from US Army ARDEC, ATTN: AMSRD-AAR-AIS-T, Picatinny, NJ 07806-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E 1742	-	Standard Practice for Radiographic Examination

(Copies of ASTM standards are available online from http://www.astm.org/ and from American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

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2.4 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subject to first article inspection in accordance with 4.2.

3.2 Materials, components and assemblies

3.2.1 Disc. Materials components and assemblies shall comply with all requirements specified on drawing 9238840.

3.2.2 Closure. Materials components and assemblies shall comply with all requirements specified on drawing 9238841.

3.2.3 Burster tube. Materials components and assemblies shall comply with all requirements specified on drawing 9238842.

3.2.4. Base plug. Materials components and assemblies shall comply with all requirements specified on drawing 9238843.

3.2.5. Projectile burster M54A1 assembly. Materials components and assemblies shall comply with all requirements specified on drawing 9238839 and this specification.

3.3 Ammunition lot numbering. Ammunition lot numbering shall be assigned in accordance with MIL-STD-1168.

3.4 Burster tube loading. The interior surface of the burster tube shall be dry when the high explosive is poured into the burster tube. Solid explosives shall not be present in the burster tube before loading or be added to the burster tube during pouring or cooling. No explosive scrap, clean or otherwise, shall be remelted or reused. Riser scrap, scrap from floor sweepings, ventilation and suction apparatus, or scrap from drilling operations shall not be reused or remelted.

3.5 Explosive composition. The explosive composition shall be prepared in accordance with drawing 9241806. Composition B-5 shall conform to the requirements specified in Table I, when tested in accordance with paragraph 4.4.2. Composition B-5 is prepared by the addition of TNT to molten Cyclotol, Type I (75/25). The elastomer and an excess of the Glyceride of 12-hydroxystearic acid are added separately and the mixture is stirred until uniform.

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TABLE I. Composition B-5 properties and requirements

Property	Requirement
RDX%	67.0 +/- 10.0%
TNT%	30.5 +/- 9.5%
Resin%	0.7 to 4.0

3.6 Explosive loading radiographic requirements.

3.6.1 Cavitation (base). There shall be no single cavity having a projected length of more than 1/8 inch or sum total of cavities of more than 1/64 square inch in the area from the top of the base plug to one inch above the top of the plug. There shall be no cracks or annular rings in the charge in this area except that cracks less than 1/32 inch wide shall be disregarded.

3.6.2 Cavitation (top) There shall be no single cavity having a projected length of more than 3/8 inch or a sum total of cavities of more than 3/8 square inch and there shall be no more than three transverse cracks over 1/32 inch wide in the area from the top of charge to one inch from the top of the base plug. Cracks less than 1/32 inches wide in this area shall be disregarded.

3.6.3 Porosity. From the top of the base plug to one inch above the top of the plug, porous areas shall not exceed a projected length of 0.47 inches or a projected area of 0.16 square inch. From one inch above the top of the base plug to the top of the charge, porous areas shall not exceed a projected length of 9.5 inches or a projected total area of 4.75 square inches.

3.6.4 Void between charge and base plug. There shall be no void greater than 1/32 of an inch between the charge face and the face of the base plug.

3.7 Workmanship. All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be clean and free of burrs, sharp edges, unblended radii, cracks, surface defects, chips, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any parts nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

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4. VERIFICATION

TABLE II Requirement / verification cross-reference matrix

Requirement / verification cross-reference matrix

CLASSES OF VERIFICATION

N/A – NOT APPLICABLE
1 - ANALYSIS
2 – DEMONSTRATION
3 - EXAMINATION
4 - TEST

METHOD OF VERIFICATION

A - FIRST ARTICLE
B - ACCEPTANCE

SECTION 3 REQUIREMENTS		VERIFICATION METHOD					VERIFICATION CLASS		SECTION 4 VERIFICATION
Ph.	Description	N/A	1	2	3	4	A	B	
3.1	First article				X	X	X		4.2
3.2	Materials, components ands assemblies	X							
3.2.1	Disc				X		X	X	4.3.2.1
3.2.2	Closure				X		X	X	4.3.2.2
3.2.3	Burster tube				X		X	X	4.3.2.4
3.2.4	Base plug				X		X	X	4.3.2.3
3.2.5	Projectile burster M54A1				X	X	X	X	4.3.2.6
3.3	Ammunition lot number				X		X	X	4.3.2.6
3.4	Burster tube loading				X		X	X	4.3.2.5
3.5	Explosive composition					X	X	X	4.3.2.6/4.4.2
3.6	Explosive loading radiography				X		X	X	4.3.2.6/4.4.1
3.6.1	Cavitation base				X		X	X	4.3.2.6
3.6.2	Cavitation top				X		X	X	4.3.2.6
3.6.3	Porosity				X		X	X	4.3.2.6
3.6.4	Void between charge and base plug				X		X	X	4.3.2.6
3.8	Workmanship				X		X	X	4.3.2.1-4.3.2.6

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4.1 Classification of verification.

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection.

4.2.1 First article verification. When specified, a sample of the cartridge and its components shall be subjected to First Article Verification in accordance with Table IV.

4.2.2 First article quantity. First article verification shall be performed on the quantity of item indicated in Table III.

TABLE III - First article verification quantities

Part Description	Drawing	Quantity
PROJECTILE BURSTER M54A1, ASSEMBLY	9238839	25
PROJECTILE BURSTER PRIOR TO ASSEMBLY FOR COMPOSITIONAL ANNALYSIS	9238839	10
DISC	9238840	25
CLOSURE	9238841	25
BURSTER TUBE	9238842	25
BASE PLUG	9238843	25

4.2.2.1 Comp B5 explosive composition. The assemblies tested for explosive composition shall be a representative sample, per MIL-STD-1916 (Representative (stratified) sampling), from the explosive loading process (kettle mix, pour, and cooling), conducted at full production volume and rate. The sample shall be selected to represent the pours from the kettle mix and shall include at least one assembly each from the first and last pour from that kettle. Sample assemblies shall be completely cured prior to explosive composition testing.

4.2.3 Rejection. If any item fails to comply with requirements, the first article sample shall be rejected.

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TABLE IV - First article inspection

	NUMBER OF FAT SAMPLES	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
DISC			
Dwg. 9238840			
Examination for Defects	25	3.2	4.3.2.1
CLOSURE			
Dwg. 9238841			
Examination for Defects	25	3.2	4.3.2.2
BASE PLUG			
Dwg. 9238843			
Examination for Defects	25	3.2	4.3.2.3
BURSTER TUBE			
Dwg. 9238842			
Examination for Defects (prior to coating)	25	3.2	4.3.2.4
Examination for Defects	25	3.2	4.3.2.5
PROJECTILE BURSTER			
M54A1, ASSEMBLY Dwg. 9238839			
Examination for Defects	25	3.2	4.3.2.6
Comp B5 Explosive Composition	10	3.6	4.3.2.6 Note 1 /4.4.2

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4.3 Conformance inspection.

4.3.1 Lot formation. Lot formation shall be in accordance with lot formation requirement of MIL-STD-1916 paragraph “Formation and identification of lots or batches”. In addition, each lot of projectile burster cases shall contain:

a. Cyclotol of one type from one manufacturer.
b. TNT of one grade from one manufacturer.
c. Polyurethane elastomer prepolymer from one manufacturer.
d. Glyceride of 12 – hydroxystearic acid from one manufacturer.
e. Tubes and components of one lot interfix number from one manufacturer.

4.3.2 Classification of characteristics. Conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. When cited herein, attributes sampling inspection shall be conducted in accordance with MIL-STD-1916, using the inspection levels cited in the Classification of Characteristic paragraphs. Definitions of Critical, Major and Minor shall be as defined in MIL-STD-1916. Acceptance criteria shall be in accordance with MIL-STD-1916. Unless automatic AIE is used for final inspection, Critical characteristics shall be inspected 100% followed by a sampling inspection in accordance with Verification Level VII of MIL-STD-1916.

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.1	TITLE Disc		SHEET 1 OF 1	DRAWING NUMBER 9238840 NEXT HIGHER ASSEMBLY 9238839
CLASSIFICATION	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**
Critical	None defined			
Major				
101	Diameter	Level IV	3.2	AIE
102	Thickness	Level IV	3.2	AIE
Minor				
201	Evidence of poor workmanship	Level II	3.7	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.2	TITLE Closure		SHEET 1 OF 1	DRAWING NUMBER 9238841 NEXT HIGHER ASSEMBLY 9238839
CLASSIFICATION	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**
Critical	None defined			
Major				
101	Outside diameter	Level IV	3.2	AIE
102	Inside diameter	Level IV	3.2	AIE
103	Thickness	Level IV	3.2	AIE
Minor				
201	Evidence of poor workmanship	Level II	3.7	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.3	TITLE Base plug		SHEET 1 OF 1	DRAWING NUMBER 9238843 NEXT HIGHER ASSEMBLY 9238839
CLASSIFICATION	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**
Critical	None defined			
Major				
101	Large diameter, min.	Level IV	3.2	AIE
102	Small diameter, min.	Level IV	3.2	AIE
103	Total length	Level IV	3.2	AIE
104	Distance to shoulder of large diameter	Level IV	3.2	AIE
Minor				
201	Chamfer missing	Level II	3.2	Visual
202	Evidence of poor workmanship	Level II	3.7	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.4	**TITLE** Burster tube (prior to coating)		**SHEET** 1 **OF** 1	**DRAWING NUMBER** 9238842
				NEXT HIGHER ASSEMBLY 9238839
CLASSIFICATION	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**
Critical	None defined			
Major				
101	Total length, min.	Level IV	3.2	AIE
102	Diameter, min.	Level IV	3.2	AIE
103	Wall thickness	Level IV	3.2	AIE
Minor				
204	Evidence of poor workmanship	Level II	3.7	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.5	**TITLE** Burster tube		**SHEET** 1 **OF** 1	**DRAWING NUMBER** 9238842
				NEXT HIGHER ASSEMBLY 9238839
CLASSIFICATION	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**
Critical	None defined			
Major				
101	Interior or exterior surface not completely coated or dry	100%	3.5	Visual/Tactile
102	Foreign matter in interior	100%	3.5	Visual/Tactile
103	Solid explosives in interior	100 %	3.5	Visual/Tactile
Minor				
201	Evidence of poor workmanship	Level II	3.7	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.6	**TITLE** Projectile burster M51A1 assembly		**SHEET** 1 **OF** 1	**DRAWING NUMBER** 9238839
				NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Radiographic inspection	100%	3.7	4.4.1
2	Base plug missing	100%	3.2	Visual
3	Burster tube cracked	100%	3.2	Visual
4	Explosive charge loose in container	100%	3.2	Manual
5	Up arrow points in wrong direction	100%	3.2	Visual
6	Closure missing	100%	3.2	Visual
Major				
101	Total length	Level IV	3.2	AIE
102	Chamber gage	Level IV	3.2	AIE
103	Explosive composition	4.4.2	3.5	4.4.2
Minor				
201	Closing disc ruptured	Level II	3.2	Visual
202	Closing disc or base plug incompletely sealed	Level II	3.2	Visual
203	Ammunition lot numbering incorrect or illegible	Level II	3.2-3.3	Visual
204	Depth of explosive from top of tube	Level II	3.2	AIE
205	Evidence of poor workmanship	Level II	3.7	Visual
				Visual

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4.4 Testing. Testing is described in the Conformance Inspection Tables and the following:

4.4.1 Radiographic examination for explosive loading defects (see 3.6). The test shall be performed as specified in 4.4.1.1. Cavitation defects are classified as follows:

Critical: Any explosive loading defect within one inch of the top of the base plug (see 3.6.1, 3.6.3, and 3.6.4).

Major: Any explosive loading defect more than one inch from the top of the base plug (see 3.6.2 and 3.6.3).

4.4.1.1 Determination of cavitation. Radiographic equipment, operations and procedures shall be qualified in accordance ASTM E 1742. The sensitivity of the radiographic technique shall be 2.0 percent and the minimum photographic density of the film used shall be no less than 1.50 at every point within the projected area of explosive. The radiographic beam shall be located in a plane which is perpendicular to the longitudinal axis of the burster and tangent to the base. The outline of the explosive fill shall be clearly defined at all points. Examination of the radiograph shall be made to determine compliance with applicable requirements. No images of lead masks, containers, fixtures, identifying marks, or other extraneous features should project into the region of explosive under inspection, with the exception of specified Image Quality Indicators. Such screening should be free of any radiographically observable features in the projected area under inspection. The maximum density shall be no greater than required to permit adequate inspection over the entire area of interest. All defect-like features in the area of interest on a radiographic image shall be treated as defects unless identified as extraneous artifacts, in which case the burster shall be x-rayed again. All additional radiographs taken for defect conditions shall be identified on the film as retakes and stored in a manner which is traceable to the original radiograph.

4.4.1.2 Check of sensitivity of radiographic technique. The sensitivity of radiographic technique shall be determined by means of Image Quality Indicators (IQI's) fabricated as specified herein. When radiographing a number of similar bursters the frequency of the sensitivity check shall be no less than one in each fifty radiographs, and on a rotating ring, the Image Quality Indicators shall be used at one station for each complete rotation of the ring. The sensitivity of the radiographic technique shall be considered satisfactory when the image of the Image Quality Indicators, including the 2T holes, are perceptible in accordance with ASTM E 1742. The Image Quality Indicators shall be made of steel having an approximate specific gravity of 7.8. For further details on construction of the Image Quality Indicators see ASTM E 1742.

4.4.1.3 Check of photographic density of film. The photographic density of the film shall be determined on the film in the region of minimum film density within the projected area of the explosive charge. Film density shall be checked each hour of operation or fraction thereof, for each film processor. Greater frequency shall be required for any period during which difficulty is encountered in establishing the required densities. Photographic densities shall be measured with a diffuse density, photoelectric densitometer providing direct display of numerical density units. Accuracy of densitometer shall be maintained by calibrating with a

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photographic step tablet at the start, middle, and end of each shift and at any time that the densitometer is turned off and started prior to taking any readings. Calibration shall be made at a step tablet density level within .5D of that being measured on the radiographic film. Each point of density measurement on the radiographs shall be identified by a circumscribed circle (made with a suitable pen or pencil) of no more than 1/4 inch in diameter. All photographic step tablets used in normal operation shall be checked for accuracy monthly by calibrating the densitometer with a reference photographic step tablet. Reference tablet shall be kept available, but shall be protected from wear and damage and shall not be used in daily operations. Any photographic step tablet having severe surface damage or showing a deviation from previous calibrations by .05D or more shall be replaced with a new, unused step tablet.

4.4.2 Composition test. Eight burster tubes shall be randomly selected from each inspection lot. An inspection lot shall be one week's production or 1,000 bursters, whichever is less. In addition, one burster tube shall be randomly selected from the first 20 burster tubes poured from the first kettle of the inspection lot and one burster tube shall be randomly selected from the last 20 burster tubes poured from the last kettle of the inspection lot. Two samples of composition B-5 shall be taken from each of the ten burster tubes per inspection lot. One sample shall be taken from an area located one inch from the top of each tube. A second sample shall be taken from an area located one inch from the bottom of the same tube. This procedure shall yield 20 samples (two separate samples for each burster tube). If any one sample fails to comply with any of the requirements specified, the burster tube inspection lot shall be rejected. The tests shall be performed as specified in 4.4.2.1 – 4.4.2.6. The resin in the sample explosive compositions shall be fully cured. A minimum cure of 7 days at room temperature or 48 hours at + 140 degrees F is recommended.

4.4.2.1 Composition test scope. This gravimetric method is designed for the analysis of the three components TNT, RDX, (with HMX) and cured resin. It consists of solvent extractions of Composition B-5 samples using 1,2 dichloroethane saturated with RDX and HMX for the TNT (filtrate determined for TNT by UV spectroscopy) and acetonitrile for the cured resin. The RDX content is determined by weight differences. A polypropylene (PPE) filter medium will be used in this gravimetric method.

4.4.2.2 Specimen. Samples of Composition B-5 shall be ground in a Wiley Mill using a 20 mesh screen. After grinding, mix the sample thoroughly to obtain a uniform representative sample.

4.4.2.3 Apparatus

a. Wiley Mill with 20 mesh screen.
b. Vacuum filtration system (consisting of pump, hosing and cold traps).
c. Kontes 1000 ml filtering dome or equivalent.
d. A Milipore 47 mm filtration system (top funnel coated with Teflon on the bottom surface and a bottom metal screen filter support).
e. Conical 50 ml graduated cylinder.
f. Beakers (sizes 50ml, 100ml).
g. 47mm No. 3 filter paper.
h. 47mm polypropylene (PPE) filters with 5 um porosity (Millipore).

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i. Aluminum dishes.
j. Magnetic stirrers with bar (2”).
k. Analytical balance (weigh to nearest 0.1mg).
l. 50ml volumetric flasks with stoppers.
m. Plastic funnel for 50ml flask.
n. UV-VIS Spectrophotometer (equivalent to HP) with 10ml UV cell with 9mm spacer or 1mm UV cell with holder for 10mm space.
o. Pasteur pipettes with rubber bulbs.
p. Sonicator bath with cooling.
q. 100 degree C drying oven.

4.4.2.4 Materials

a. 1,2 dichloroethane (DCE).
b. RDX.
c. HMX.
d. TNT (with lot number; used as spectrophotometric standards).
e. Acetonitrile.

4.4.2.5 TNT determination

a. Prepare RDX/HMX saturated 1,2 dichloroethane (DCE). Add excess RDX and HMX to 4 liters of DCE and stir with magnetic stirrer (2” stirring bar) for 2 hours. Allow the saturated solution to settle and filter with Millipore filtering system and No. 3 paper filter.
b. Set up filtration system using Kontes filtering dome and conical graduated cylinder. Insert funnel support.
c. Weigh polypropylene filter in labeled aluminum disk.
d. Place a 47mm filter paper on top of funnel support.
e. Place weighed PPE filter on top of filter paper.
f. Clamp funnel onto funnel support.
g. Weigh approximately 1.5g of ground composition B-5 sample into the 50ml beaker.
h. Add 30ml saturated RDX/HMX DCE solvent. Stir and sonicate in cold water (ice may be used) for 40 seconds.
i. Turn vacuum pump on. Close valve on filtering bell.
j. Filter beaker contents thru filtering apparatus. Wash contents of beaker into filtering apparatus using saturated RDX/HMX DCE with a Teflon wash bottle. Cap volumetric flask and allow to stand 1 hour before filling to mark and performing the TNT analysis.
k. Unclamp funnel from funnel support.
l. With wash bottle rinse edges of PPE filter into graduated cylinder.
m. Turn off vacuum by releasing valve on filtering bell.
n. Use forceps to transfer PPE filter to the aluminum dish.

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Place PPE/aluminum dish in a 100 degree C drying oven for one half hour.

o. Cool PPE/aluminum dish for 20 minutes before weighing.

p. Determine percentage of RDX/HMX with Resin:

$$\%\text{RDX/HMX and Resin} = \frac{100\ (A-B)}{\text{Weight of Sample}}$$

A=PPE filter, aluminum dish and filtrate after drying.

B= PPE filter and aluminum dish.

q. UV Analysis of TNT Procedure:

r. Warm up UV spectrophotometer, and measure absorbance at 374 nm.

s. For a programmable UV instrument, use a 5 second integration time.

t. Prepare TNT standards of 0.4g, 0.5g and 0.6g in 50ml flasks in unsaturated DCE.

u. Run solvent blank using 10mm cell with spacer (or 1mm cell). Fill cell with a Pasteur pipette. Wipe cell faces with soft tissue before reading.

v. Next run three TNT standard solutions. Check linearity of calibration curve. Correlation coefficient should be close to 1.000.

w. Scan the filtrate solution to determine grams of TNT/50ml.

$$\%\text{TNT} = \frac{100\ (\text{g of TNT per 50ml by UV})}{\text{Weight of Sample}}$$

Note: TNT content by UV can be confirmed by comparing it to the DCE extracted weight. If the UV content is significantly greater, the sample may be contaminated.

4.4.2.6 Determination of cured resin and RDX/HMX

a. Connect the Kontes filtering dome to vacuum. Insert the funnel support with tip inserted into the beaker. Place a piece of filter paper on the funnel support.

b. Weigh a PPE filter in a labeled aluminum dish.

c. Place the PPE filter on top of the filter paper. Clamp the filter funnel in place.

d. Add 2 grams of the ground composition B-5 sample to a 100 ml beaker.

e. Add 60-70ml of acetonitrile. Sonicate for 5-6 minutes.

f. Turn on vacuum and close valve on Kontes dome.

g. Filter beaker contents. Rinse with acetonitrile with vacuum on using a solvent resistant wash bottle.

h. Unclamp and remove filter funnel. Rinse edges of filter with acetonitrile.

i. Turn off vacuum and release valve on Kontes dome.

j. Use forceps to remove PPE filter and place in an aluminum pan to dry in 100

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degree C oven for 5 minutes.

k. Allow pan to cool for 3 minutes before weighing. Complete the calculations below.

$$\% \text{ Resin} = \frac{100(C-D)}{\text{Weight of Sample}}$$

C= Weight of PPE filter, aluminum pan and filtrate after drying.
D= Weight of PPE filter and aluminum pan.
% RDX/HMX=%RDX/HMX and Resin - %Resin

4.4.2.7. Alternate acceptance method. After all assemblies from five consecutive inspection lots have met all requirements for explosive composition, a request for an alternate acceptance method based on kettle composition may be submitted. (See 6.1)

5 PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (See 6.2). When actual packaging of materiel is to be performed by DOD personnel, or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The components and assemblies covered by this specification are intended for use in the 155MM, WP, M110 Projectile. The M54A1 Projectile Burster covered by this specification is military unique because it is used for 155MM howitzer ammunition only and has no commercial application.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Packaging requirements. (See 5.1)
c. Requirements for submission of first article sample.
d. Certificate of Conformance for each lot or shipment of product.
e. Provisions for Government furnished material.
f. Requirements for submission of Contractor acceptance inspection equipment designs. (See 6.4)
g. Requirements for submission of zinc phosphate coating procedure. (See 6.5)

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h. Requirements for Certification of nondestructive test personnel. (See 6.6)
i. Requirements for submission of radiographic procedure. (See 6.7)
j. Requirements for submission of alternate explosive composition acceptance method based on kettle composition. (See 6.9)

6.3 Ammunition lot numbering. Ammunition lot numbers to be assigned in accordance with MIL-STD-1168.

6.4 Submission of contractor acceptance inspection equipment (AIE) designs for approval. Submit copies of designs as required to: Commander U.S. Army ARDEC, ATTN: AMSRD-AAR-QEM-A Picatinny, NJ 07806-5000 This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.5 Zinc phosphate coating procedure. Unless otherwise specified by the procuring agency, details of the proposed procedure including chemicals, equipment, and control tests to be used by the contractor, are to be submitted in writing to the contracting office. Written approval must be granted prior to the commencement of production. (See TT-C-490)

6.6 Certification of nondestructive test (NDT) personnel. All personnel operating or calibrating NDT equipment to be qualified and certified in accordance with NAS-410. Qualification to other standards must be submitted to and approved by Commander U.S. Army ARDEC, ATTN: AMSRD-AAR-QEM-A Picatinny, NJ 07806-5000.

6.7 Radiographic procedures. Submit radiographic procedures, standards and sample radiographs to: Commander U.S. Army ARDEC, ATTN: AMSRD-AAR-QEM-A Picatinny, NJ 07806-5000 for review and approval.

6.7.1 Radiographic film. Only a film type and vendor that has been approved by the Government is to be used for radiographing this projectile. The final choice of film type must be determined for the particular radiographic facility to be used and will be based on achieving adequate sensitivity for both the IQI standard and inspected projectiles. A substitute item is not to be used before prior testing and approval by the Government

6.7.2 Non-film methods of radiographic inspection. No non-film methods of radiographic inspection should be employed without prior review and approval by ARDEC. Qualification to be in accordance with ASTM E-1411 (Standard Practice for Qualification of Radioscopic Systems) and an approved Qualification Plan. System performance and implementation to comply with ASTM E-1255 (Standard Practice for Radioscopy). Qualification must include a demonstration that the specific non-film system proposed for production inspection detects or discloses all types and sizes of defects requiring radiographic examination.

6.8 Chromous chloride solution. Method for the preparation of 0.2 N chromous chloride solution may be found in Picatinny Arsenal Technical Memorandum No. ACS-360, "Determination of Nitrogenous Compounds of Ordnance Interest by Chromous Chloride Reduction (1) Compounds Containing Nitro and Nitramine Groups" by Charles C. Jamison dated

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May 1960 which may be obtained from Picatinny, New Jersey 07806-5000 ATTN: AMSRD-AAR-QEM-A.

6.9. Alternate explosive composition acceptance method based on kettle composition. Request for Government approval of an alternate explosive composition acceptance method based on kettle composition should be in accordance with the requirements of MIL-STD-1916, paragraph 4.1 to 4.1.4 and 5.1 to 5.1.4.3. The request should include, as a minimum, data from kettle samples taken just prior to the first tube poured and just after the final tube poured, and from tube samples taken from among the first and last 25 poured from each kettle. The request should be made to the design agency; ARDEC, Attn: AMSRD-AAR-QEM-A, Picatinny, NJ 07806-5000.

6.10 Critical characteristic controls. All production contracts must contain requirements for handling critical defects. As part of all proposals, suppliers should be asked to describe policies, procedures and controls for all operations associated with safety characteristics, how they are documented and maintained under the supplier's integrated management system.

6.11 Definitions.

a. Acceptance Inspection Equipment (AIE). Any measuring device that is traceable to the national or international standard used to assure conformance of material to the contract requirements.

6.12 Subject term (key word) listing.

Ammunition
Artillery
First Article

Custodian:
Army – AR

Preparing Activity:
Army – AR
(Project 1320-2006-006)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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